

MANAGEMENT OF ACNE AND SKIN INFECTIONS

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ACNE VULGARIS

Introduction

Acne vulgaris is a chronic inflammatory disease of the pilosebaceous unit characterised by the formation of comedones, erythematous papules, pustules, and/or nodules (i.e. pseudocysts) that can be accompanied by scarring.

Management of Acne Vulgaris

1. Patient Education

- Gentle soap-free pH-balanced cleanser
- Non-comedogenic or oil-free cosmetics
- Acne-specific moisturiser and sunscreen
- Low glycaemic index diet and avoid dairy product

2. Determine the severity of acne vulgaris using Comprehensive Acne Severity Scale (CASS)

GRADE		DESCRIPTION
Clear	0	No lesions to barely noticeable ones; very few scattered comedones and papules
Almost clear	1	Hardly visible from 2.5 meters away; a few scattered comedones and a few small papules; and very few pustules, comedones and papules
Mild	2	Easily recognizable; less than half of the affected area is involved; many comedones, papules and pustules
Moderate	3	More than half of the affected area is involved; numerous comedones, papules and pustules
Severe	4	Entire area is involved; covered with comedones; numerous pustules and papules; a few nodules and cysts
Very severe	5	Highly inflammatory acne covering the affected area, nodules, and cysts present

3. Prescribe the appropriate medications for the patient after the assessment of the severity of acne vulgaris using CASS.

Disease Severity	Type of Therapy	Agent	Side Effect / Comment
Mild Acne	Topical	Morning: - Benzoyl peroxide 5% gel OM OR - Acne cream/lotion OM ± Clindamycin 1% external solution or gel OD	Burning, erythema, stinging and pruritus. Benzoyl peroxide can cause bleaching of hair and clothes. Inform patients to use skin protection e.g., sunscreen and minimize prolonged exposure to sun. Acne cream to facial lesions and acne lotion to back lesions. Mild peeling of skin and slight reddening is normal. If irritation is excessive, stop treatment and consult the doctor. Topical antibiotic monotherapy is not recommended as it is associated with rapid antibiotic resistance.
		Night: Adapalene 0.1% ON (\$\$)	May produce irritant contact dermatitis and photosensitivity (use of sunscreens recommended) Strategies for improving tolerability of topical agents: - Initiating every-other-day or every 3 daily dosing with titration upward slowly as needed - Concomitant use of moisturizer; avoid astringents and toners
Moderate Acne	Topical + Systemic	Same treatment as mild acne + Doxycycline For those who cannot use doxycycline: Erythromycin Second line: Minocycline (Not available in NUP)	Add on topical retinoids or benzoyl peroxide to oral antibiotics. Oral antibiotics (should not exceed 3 – 4 months to minimize the risk of resistance): should not be used as single agent or with another topical antibiotic and should be combined with other recommended topical agents. • Can consider starting at Doxycycline 100mg BD OR Erythromycin 500mg BD. • Review 4 – 6 weeks post initiation for improvement with a view to taper to Doxycycline 100 mg OD OR Erythromycin 250mg BD. Please take note that treatment for a minimum duration of 4 – 6 weeks is generally required to see a clinical improvement. The dosage is maintained until clearance, after which it is gradually decreased. Some patients require a low dose maintenance treatment for 1-2 months to prevent relapse. • Stop oral antibiotics when inflammatory lesions clear and ensure that patients adhere to a topical regime.

Disease Severity	Type of Therapy	Agent	Side Effect / Comment
			If no improvement despite oral antibiotics, consider oral contraceptives (for females) or dermatology referral. May cause gastrointestinal tract upset, to take with food. Doxycycline can cause photosensitivity. Sun protection is advised.
Severe Acne	Systemic	Oral Isotretinoin (*Not Available in NUP)	Refer to Dermatologist

4. Differential diagnosis of acne vulgaris

Diagnosis	Important Factors	Location	Clinical Features
Rosacea	Slow onset aggravated by cold, alcohol, hot foods, stress; unknown aetiology	Central face	Erythema, telangiectasias, papules / pustule; can present with rhinophyma or chronic eye inflammation
Perioral dermatitis	Sometimes associated with prolonged use of high-potency topical steroids	Chin, perioral and nasolabial folds	Papules, pustules, and erythema confined to the chin and nasolabial folds with sparing of the area directly adjacent to the vermilion border
Gram-negative folliculitis	Can be seen with long-term antibiotic therapy	(a) Nose and mouth areas (common) (b) Neck (uncommon)	(a) Superficial pustules (b) Large nodules
Steroid acne	Associated with oral corticosteroid therapy	Chest, back, upper arms, face	Small, monomorphic papules, pustules or closed comedones
Pityrosporum folliculitis	Increase during hot, humid weather or increased sweating	Chest, back	Absence of comedones and a suspicion history, e.g. in a newly conscripted national serviceman.

5. Specialist Consultation

- Severe acne
- Nodulocystic acne
- Unsatisfactory response to treatment after 2 months of oral antibiotics
- Acne scars

SKIN INFECTIONS

Management of Common Viral Infections

1. Varicella (Chicken Pox)

- Avoid scratching.
- Medications for adult: Oral Acyclovir 800 mg 5X/day for 1 week (renal adjusted dose is required for known CKD patients). Refer to Paediatric dose chart for paediatric dosing of 5-days duration if oral acyclovir is assessed to be required. Consult pharmacist for further information.
- Medical Leave for 10 - 14 days.
- Refer severe complicated cases & pregnant women for assessment.

2. Herpes Zoster

- Uncomplicated herpes zoster – treat with Acyclovir 800mg 5x/day for 1 week (renal adjusted dose is required for known CKD patients), with adequate analgesia.
- Severe multidermatomal or disseminated zoster cases should be referred to ED for admission.
- Post Herpetic Neuralgia:
 - Acute phase: WHO analgesia ladder;
 - Persistent pain after 1 week: consider Gabapentin / Tricyclic anti-depressants.
- Refer ophthalmologist if dermatomal involvement of V1 of trigeminal nerve or refer neurologist if post herpetic neuralgia not responsive to analgesia.

3. Hand Foot Mouth Disease

- Treatment: Supportive care, good hygiene practices.
- Medical Leave for 7 - 10 days.
- Refer A&E if poor oral intake.
- Memo for fitness for school if required:
 - No fever
 - No oral ulcers
 - No blisters in hands / arms / feet / legs / buttocks

4. Viral Warts

- Medication: Topical Salicylic Acid Lotion ON up to 12 weeks (Not to be used for face or genital warts).
- Contraindications: Children $\leq$ 2 years old, patients with diabetes or impaired blood circulation. To consult senior doctor or clinic dermatologists.
- Refer if:
 - Recalcitrant warts
 - Unsatisfactory response after 2 months of topical keratolytic agents at primary care level
 - Periungual or subungual warts
 - Facial warts
 - Cutaneous horns or ulcerated lesions
 - Immunosuppressed patients (Renal transplant, HIV)

5. Viral Exanthem

- Expectant management. Resolve in few days - 3 weeks

6. Herpes Simplex

- First line treatment:
 - Localised infections such as cold sores at angles of mouth: PO Acyclovir 400mg TDS x 5 – 10 days
- Second line treatment: Valacyclovir (*Not Available in NUP)

Management of Common Bacterial Infections

1. Folliculitis, Furuncle, Impetigo

- First line treatment:
 - For non-infections, consider non-antibiotic alternatives:
Creams: Chlorhexidine 1% cream
Solution: Denatured alcohol 70%, Potassium permanganate 0.1% solution, Chlorhexidine 0.05% solution.
 - Mild, localized: Topical Fusidic acid 2% cream Q12H 1 week OR Tetracycline 3% ointment Q12H 1 week.

2. Abscess, Cellulitis

- First line treatment:
 - Cephalexin 500mg Q8H x 5 days, OR Cloxacillin 500mg Q6H x 5 days
- Second line treatment:
 - Clindamycin 300-450mg Q6H x 5 days
- Elevate affected area.
For abscess, drainage is recommended with hygiene and dressing advice.
Consult pharmacist for renal adjustments as required.

Management of Common Fungal Infections

1. Tinea corporis / cruris / pedis

- First line treatment:
 - Send for fungus smear.
 - Topical Miconazole 2% cream BD/Clotrimazole 1% BD: advise to apply on affected areas with 2cm normal borders, to continue for 1 week after symptoms resolve.
 - ± Ketoconazole 2% shampoo OD as wash to affected areas – leave on 5 minutes before rinsing.
 - ± Clotrimazole 1% powder BD to affected areas.
- Second line treatment (consider if BSA > 10% or failure to clear with topical treatment):
 - PO Itraconazole = 200mg OD 1 week or 100mg BD 2 weeks (there is variation in practice) – consider baseline LFT and evaluate for drug interactions.
 - Exclude co-existent diabetes mellitus or underlying immunosuppression.

2. Tinea Versicolor

- First line treatment:
 - Send for fungus smear.
 - Ketoconazole 2% shampoo as wash to affected areas – leave on 5 minutes before rinsing.
 - Ketoconazole 2% cream BD x 14 days or until clinical resolution.
 - Other options available: Selenium sulfide shampoo.
- Second line treatment (lack of response to topicals/relapses):
 - PO Itraconazole 200mg OD x 5 – 7 days – consider LFT and evaluate for drug interactions.
 - Exclude co-existent diabetes mellitus or underlying immunosuppression.

3. Pityrosporum folliculitis

- First line treatment:
 - Ketoconazole 2% shampoo EOD as wash to affected areas – leave on 5 minutes before rinsing.
 - Ketoconazole 2% cream BD x 4 weeks or until clinical resolution.
- Second line treatment (lack of response to topicals / relapses):
 - PO Itraconazole 200mg OD x 1 – 3 weeks – consider baseline LFT and evaluate for drug interactions.

4. Onychomycosis

- First line treatment:
 - Clotrimazole 1% lotion BD to affected nails until clinical resolution; moderate expectations as improvement may require months.
- Refer to dermatologist if considering for oral antifungal treatment (poor response to topical treatment, 4 or more nails affected)
 - Appropriate fungal cultures should be taken before instituting oral therapy. Therefore, oral antifungal treatment is best initiated by dermatologists in our setting.
 - Oral antifungals are generally contraindicated in liver disease / congestive cardiac failure.

Management of Common Parasite Infections

1. Scabies

- To be suspected in all patients presented with presenting with pruritic scaly papules, especially those involving web spaces, flexures, anogenital areas.
- Enquire about nursing home contacts or other close contacts with similar symptoms.
- First line treatment:
 - May treat empirically with topical Malathion 0.5% (can be used for children > 6 months old) lotion for all suspected cases of scabies.
 - Apply to all areas of the body from neck down. Wash off after 24 hours. Repeat in 1 week (be careful of skin irritation risk with malathion).
 - Remember to treat all close contacts at the same time to avoid reinfestation. Advise to machine wash and treat clothing/linen with iron or dryer (at least 60° Celsius).
 - Non-sedating antihistamines (e.g. loratadine or cetirizine) OM and sedating antihistamine ON (e.g., chlorpheniramine or hydroxyzine)
 - Watch for secondary bacterial infection as extensive excoriations or evidence of impetiginisation ('golden crusting') may warrant a course of antibiotics.
 - Post-scabietic itch may occur and may be treated with topical corticosteroids after adequate treatment with malathion.
- Second line treatment or for Children < 6 months old: Refer to Dermatologist for Permethrin (Not available in NUP).

2. Lice

- Diagnosis is made by direct visualisation of adult lice or nits (to check seams of clothing as well). This can affect the hair / scalp, body, or pubic area.
- First line treatment:
 - Apply Malathion 0.5% lotion on hair and scalp, trunk and/or pubic area for 12 hours before washing off. A single application is sufficient but if live lice are visualized 7-9 days after treatment, a second treatment should be performed. Beware of local skin irritation. Not recommended for infants below the age of 6 months.
 - Children can return to school immediately after completion of first application of Malathion.
 - For head lice, wet combing can be done every 3 to 4 days until no live louse is found for a continuous 2 week period.
 - Close contacts should be examined and treated if infested.
 - Treatment of clothing / linen as per scabies.
 - Give non-sedating antihistamines (such as loratadine or cetirizine) during the day and sedating antihistamine at night (such as chlorpheniramine or hydroxyzine)
 - Treatment with topical corticosteroids twice a day after eradication of lice may be helpful for symptom relief.
- Second line treatment: Permethrin lotion ([Not Available in NUP](#))

REFERENCES

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